

Architecture Development Part 3: Generation of Synthea Patient and Syndromic Surveillance Data for Hospital EHRs to Simulate Disease Outbreak

A. Synthea Generation

1. Aspirus generation

```
Activities Terminal Mar 12 01:16 sat3210@uphle: ~/synthea
openjdk_04-Bit Server VM (build 17.0.13+0-Ubuntu-0ubuntu12.04, mixed mode, sharing)
(base) sat3210@uphle:~/synthea$ ./run_synthea -p 20 Michigan Calumet --config covid19
Starting a Gradle Daemon (subsequent builds will be faster)

> Task :compileJava
Note: Some input files use unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.

> Task :run
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#noProviders for further details.
Scanned 90 modules and 157 submodules.
Loading submodule modules/allergies/allergy_panel.json
Loading submodule modules/allergies/drug_allergy_incidence.json
Loading submodule modules/allergies/environmental_allergy_incidence.json
Loading submodule modules/allergies/food_allergy_incidence.json
Loading submodule modules/allergies/immunotherapy.json
Loading submodule modules/allergies/outgrow_env_allergies.json
Loading submodule modules/allergies/outgrow_food_allergies.json
Loading submodule modules/allergies/severe_allergic_reaction.json
Loading submodule modules/anemia/anemia_sub.json
Loading submodule modules/breast_cancer/chemotherapy_breast.json
Loading submodule modules/breast_cancer/hormone_diagnosis.json
Loading submodule modules/breast_cancer/hormonotherapy_breast.json
Loading submodule modules/breast_cancer/surgery_therapy_breast.json
Loading submodule modules/breast_cancer/tnm_diagnosis.json
Loading submodule modules/contraceptives/clear_contraceptive.json
Loading submodule modules/contraceptives/female_sterilization.json
Loading submodule modules/contraceptives/implant_contraceptive.json
Loading submodule modules/contraceptives/injectable_contraceptive.json
Loading submodule modules/contraceptives/intrauterine_device.json
Loading submodule modules/contraceptives/male_sterilization.json
Loading submodule modules/contraceptives/oral_contraceptive.json
Loading submodule modules/contraceptives/patch_contraceptive.json
Loading submodule modules/contraceptives/ring_contraceptive.json
Loading submodule modules/covid19/admission.json
Loading submodule modules/covid19/determine_risk.json
Loading Lookup Table: covid-19-severity-outcomes.csv
Loading Lookup Table: covid-19-survival-outcomes.csv
Loading submodule modules/covid19/diagnose_bacterial_infection.json
Loading submodule modules/covid19/diagnose_blood_clot.json
Loading submodule modules/covid19/end_outcomes.json
Loading submodule modules/covid19/end_symptoms.json
Loading submodule modules/covid19/infection_type
```

```
Activities Terminal Mar 12 01:19 sat3210@uphle: ~/synthea
Loading Lookup Table: vhd_ps.csv
Loading Lookup Table: vhd_pr.csv
Loading module modules/vhd_tricuspid.json
Loading Lookup Table: vhd_tr.csv
Loading Lookup Table: vhd_ts.csv
Loading module modules/wellness_encounters.json
Running with options:
Population: 20
Seed: 1773287867357
Provider Seed: 1773287867357
Reference Time: 1773287867357
Location: Calumet, Michigan
Min Age: 0
Max Age: 140
1 -- Wilburn655 Romaguera67 (15 y/o M) Calumet, Michigan (20843)
2 -- Fabian647 Feest103 (60 y/o M) Calumet, Michigan (82054)
3 -- Ute382 Cronin387 (24 y/o F) Calumet, Michigan (33952)
4 -- Barney639 Abernathy524 (22 y/o M) Calumet, Michigan (30677)
6 -- Prudence900 Candra395 Bogitsich202 (30 y/o F) Calumet, Michigan DECEASED (45209)
5 -- Houston994 Kilrin939 (49 y/o M) Calumet, Michigan (69330)
7 -- Brad867 Huel628 (15 y/o M) Calumet, Michigan (20795)
8 -- Frankie174 Medhurst46 (18 y/o M) Calumet, Michigan (25424)
9 -- Ray930 Bernier607 (12 y/o M) Calumet, Michigan (16865)
6 -- Marx778 Herzog843 (57 y/o F) Calumet, Michigan (82204)
11 -- Spring819 Moore224 (29 y/o F) Calumet, Michigan (41599)
10 -- Lill474 Penn1796 Schuster709 (78 y/o F) Calumet, Michigan (107806)
12 -- Saran641 Gulgowski816 (39 y/o F) Calumet, Michigan (56607)
13 -- Juliette688 Champlin946 (23 y/o F) Calumet, Michigan (32999)
15 -- Gregory545 Hilpert278 (22 y/o M) Calumet, Michigan (29601)
14 -- Huey641 Bednar518 (46 y/o M) Calumet, Michigan (62703)
17 -- Barbara869 Crooks415 (26 y/o F) Calumet, Michigan (36836)
16 -- Inge747 Lilly990 Hackett68 (61 y/o F) Calumet, Michigan (85992)
18 -- Alberto639 Barrows492 (67 y/o M) Calumet, Michigan (107958)
19 -- Gregory545 Graham902 (76 y/o M) Calumet, Michigan (105307)
20 -- Davida245 Satterfield305 (16 y/o F) Calumet, Michigan (21658)
Records: total=21, alive=20, dead=1
RNG=20
Clinician RNG=11655

You've just generated 21 patients! We're happy to know Synthea is helping you with your project.
Like what we're doing? Synthea is an open-source project, and your support helps us keep building.
Please consider making a donation (https://github.com/sponsors/synthetichealth).
```

2. Portage generation

```
Activities Terminal Mar 12 01:20 sat3210@uphile: ~/synthesia
hospitalInformation1773287867357.json
(base) sat3210@uphile:~/synthesia$ ./run_synthesia -p 20 Michigan "Houghton" --config covid19

> Task :run
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#noProviders for further details.
Scanned 90 modules and 157 submodules.
Loading submodule modules/allergies/allergy_panel.json
Loading submodule modules/allergies/drug_allergy_incidence.json
Loading submodule modules/allergies/environmental_allergy_incidence.json
Loading submodule modules/allergies/food_allergy_incidence.json
Loading submodule modules/allergies/immunotherapy.json
Loading submodule modules/allergies/outgrow_env_allergies.json
Loading submodule modules/allergies/outgrow_food_allergies.json
Loading submodule modules/allergies/severe_allergic_reaction.json
Loading submodule modules/anemia/anemia_sub.json
Loading submodule modules/breast_cancer/chemotherapy_breast.json
Loading submodule modules/breast_cancer/hormone_diagnosis.json
Loading submodule modules/breast_cancer/hormonotherapy_breast.json
Loading submodule modules/breast_cancer/surgery_therapy_breast.json
Loading submodule modules/breast_cancer/tnm_diagnosis.json
Loading submodule modules/contraceptives/clear_contraceptive.json
Loading submodule modules/contraceptives/female_sterilization.json
Loading submodule modules/contraceptives/implant_contraceptive.json
Loading submodule modules/contraceptives/injectable_contraceptive.json
Loading submodule modules/contraceptives/intrauterine_device.json
Loading submodule modules/contraceptives/male_sterilization.json
Loading submodule modules/contraceptives/oral_contraceptive.json
Loading submodule modules/contraceptives/patch_contraceptive.json
Loading submodule modules/contraceptives/ring_contraceptive.json
Loading submodule modules/covid19/admission.json
Loading submodule modules/covid19/determine_risk.json
Loading Lookup Table: covid-19-severity-outcomes.csv
Loading Lookup Table: covid-19-survival-outcomes.csv
Loading submodule modules/covid19/diagnose_bacterial_infection.json
Loading submodule modules/covid19/diagnose_blood_clot.json
Loading submodule modules/covid19/end_outcomes.json
Loading submodule modules/covid19/end_symptoms.json
Loading submodule modules/covid19/infection.json
Loading submodule modules/covid19/measurements_daily.json
Loading submodule modules/covid19/measurements_frequent.json
Loading submodule modules/covid19/measurements_vitals.json
Loading submodule modules/covid19/medications.json
```

```
Activities Terminal Mar 12 01:21 sat3210@uphile: ~/synthesia
Loading Lookup Table: vnd_ts.csv
Loading module modules/wellness_encounters.json
Running with options:
Population: 20
Seed: 1773289338987
Provider Seed: 1773289338987
Reference Time: 1773289338987
Location: Houghton, Michigan
Min Age: 0
Max Age: 140
1 -- Maria984 Mariel814 Gorczyn269 (56 y/o F) Houghton, Michigan (77737)
2 -- Sean831 Padberg411 (68 y/o M) Houghton, Michigan DECEASED (119677)
3 -- Dorine689 Feeney44 (2 y/o F) Houghton, Michigan (4329)
4 -- Angel97 Spencer878 (4 y/o F) Houghton, Michigan (6775)
5 -- Harvey63 Jacobson885 (26 y/o M) Houghton, Michigan (37092)
6 -- Grant908 Haag279 (74 y/o M) Houghton, Michigan (102984)
7 -- Shell1447 Lindsay928 Grady603 (76 y/o F) Houghton, Michigan (109037)
2 -- Mckinley734 Walsh511 (70 y/o M) Houghton, Michigan (114500)
8 -- Jewell855 Gaylord332 (57 y/o M) Houghton, Michigan (87435)
9 -- Tama137 Dorothy94 Trantow673 (66 y/o F) Houghton, Michigan (92258)
10 -- Robin66 Waters156 (78 y/o M) Houghton, Michigan DECEASED (98092)
11 -- Kathryn011 Armstrong951 (21 y/o F) Houghton, Michigan (36170)
12 -- Damlan46 Lubowitz58 (58 y/o M) Houghton, Michigan (80949)
10 -- Adolph80 Breitenberg711 (71 y/o M) Houghton, Michigan DECEASED (98991)
10 -- Faustino767 Schinner682 (76 y/o M) Houghton, Michigan (105407)
13 -- Christena299 Schroeder447 (59 y/o F) Houghton, Michigan DECEASED (90345)
14 -- Eric487 Rau926 (65 y/o M) Houghton, Michigan DECEASED (88890)
13 -- Chandra831 Schaden604 (66 y/o F) Houghton, Michigan DECEASED (92420)
14 -- Coy949 Wintheliser220 (69 y/o M) Houghton, Michigan (96316)
15 -- Loria303 Williamson769 (51 y/o F) Houghton, Michigan (70883)
13 -- Juli424 Shayla126 Bechtelar572 (81 y/o F) Houghton, Michigan (121478)
16 -- Dolores502 Rutherford999 (6 y/o F) Houghton, Michigan (9252)
17 -- Rosella779 Haley279 Mutch271 (41 y/o F) Houghton, Michigan (57853)
19 -- Derek111 Predovics34 (51 y/o M) Houghton, Michigan (69920)
18 -- Luke971 Kreiger457 (78 y/o M) Houghton, Michigan (108969)
20 -- Son314 Hermiston71 (68 y/o M) Houghton, Michigan (126077)
Records: total=26, alive=20, dead=6
RNG=20
Clinician RNG=11664

You've just generated 26 patients! We're happy to know Synthesia is helping you with your project.
Like what we're doing? Synthesia is an open-source project, and your support helps us keep building.
Please consider making a donation (https://github.com/sponsors/synthetichealth).
```

3. Baraga generation

```
Activities Terminal Mar 21, 2023 sat3210@uphie: ~/synthesia  
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/9.2.1/userguide/configuration_cache_enabling.html  
(base) sat3210@uphie:~/synthesia$ ./run_synthesia -p 20 Michtgan "Baraga" --config covid19  
  
> Task :run  
SLF4J: No SLF4J providers were found.  
SLF4J: Defaulting to no-operation (NOP) logger implementation  
SLF4J: See http://www.slf4j.org/codes.html#noProviders for further details.  
Scanned 90 modules and 157 submodules.  
Loading submodule modules/allergies/allergy_panel.json  
Loading submodule modules/allergies/drug_allergy_incidence.json  
Loading submodule modules/allergies/environmental_allergy_incidence.json  
Loading submodule modules/allergies/food_allergy_incidence.json  
Loading submodule modules/allergies/immunotherapy.json  
Loading submodule modules/allergies/outgrow_env_allergies.json  
Loading submodule modules/allergies/outgrow_food_allergies.json  
Loading submodule modules/allergies/severe_allergic_reaction.json  
Loading submodule modules/anemia/anemia_sub.json  
Loading submodule modules/breast_cancer/chemotherapy_breast.json  
Loading submodule modules/breast_cancer/hormone_diagnosis.json  
Loading submodule modules/breast_cancer/hormonetherapy_breast.json  
Loading submodule modules/breast_cancer/surgery_therapy_breast.json  
Loading submodule modules/breast_cancer/ttn_diagnosits.json  
Loading submodule modules/contraceptives/clear_contraceptive.json  
Loading submodule modules/contraceptives/female_sterilization.json  
Loading submodule modules/contraceptives/inplant_contraceptive.json  
Loading submodule modules/contraceptives/injectable_contraceptive.json  
Loading submodule modules/contraceptives/intrauterine_device.json  
Loading submodule modules/contraceptives/male_sterilization.json  
Loading submodule modules/contraceptives/oral_contraceptive.json  
Loading submodule modules/contraceptives/patch_contraceptive.json  
Loading submodule modules/contraceptives/ring_contraceptive.json  
Loading submodule modules/covid19/admisstion.json  
Loading submodule modules/covid19/determine_risk.json  
Loading Lookup Table: covid-19-severity-outcomes.csv  
Loading Lookup Table: covid-19-survival-outcomes.csv  
Loading submodule modules/covid19/diagnose_bacterial_infection.json  
Loading submodule modules/covid19/diagnose_blood_clot.json  
Loading submodule modules/covid19/end_outcomes.json  
Loading submodule modules/covid19/end_symptoms.json  
Loading submodule modules/covid19/infection.json  
Loading submodule modules/covid19/measurements_daily.json  
Loading submodule modules/covid19/measurements_frequent.json  
Loading submodule modules/covid19/measurements_vitals.json  
Loading submodule modules/covid19/medications.json
```

```
Activities Terminal Mar 12 01:23 sat3210@uphile: ~/synthea

Loading Lookup Table: vhd_tr.csv
Loading Lookup Table: vhd_ts.csv
Loading module modules/wellness_encounters.json
Running with options:
Population: 20
Seed: 1773289517020
Provider Seed: 1773289517020
Reference Time: 1773289517020
Location: Baraga, Michigan
Min Age: 0
Max Age: 140
2 -- Jeffery551 Rohan584 (39 y/o M) Baraga, Michigan (55757)
1 -- Moriah648 Jayna69 Zienann98 (47 y/o F) Baraga, Michigan (64970)
3 -- Allan98 Huei628 (36 y/o M) Baraga, Michigan (52685)
4 -- Grag740 Kuitc920 (66 y/o M) Baraga, Michigan DECEASED (102403)
5 -- Ross213 Pacocha935 (63 y/o M) Baraga, Michigan (86301)
6 -- Rory188 Lynch190 (19 y/o M) Baraga, Michigan DECEASED (26321)
6 -- Merlin721 Krelger457 (40 y/o M) Baraga, Michigan (53892)
4 -- Aurelio227 Mertz280 (67 y/o M) Baraga, Michigan (92297)
7 -- Ranae267 Purdy2 (61 y/o F) Baraga, Michigan DECEASED (85769)
8 -- Jenny686 Thora330 Homenick806 (77 y/o F) Baraga, Michigan (109350)
9 -- Jenae263 Corwin846 (16 y/o F) Baraga, Michigan DECEASED (22901)
7 -- Sofia418 Dion244 Parisian75 (75 y/o F) Baraga, Michigan (107355)
9 -- Mozella148 Runte676 (24 y/o F) Baraga, Michigan (34009)
10 -- Odell1776 Bradtke547 (23 y/o M) Baraga, Michigan (22357)
11 -- Mindy87 Towne455 (34 y/o F) Baraga, Michigan (48958)
12 -- Hubert238 Von197 (40 y/o M) Baraga, Michigan (57549)
13 -- Francine922 Hammes673 (42 y/o F) Baraga, Michigan (57640)
14 -- Alvaro283 Jones311 (47 y/o M) Baraga, Michigan (63163)
15 -- Cornelius968 Williamson769 (69 y/o M) Baraga, Michigan (94285)
17 -- Todd315 Effertz744 (2 y/o M) Baraga, Michigan (3119)
16 -- Jama227 McClure239 (55 y/o F) Baraga, Michigan (83409)
18 -- Darin74 Barton704 (22 y/o M) Baraga, Michigan (30242)
19 -- Erich201 Trantow673 (32 y/o M) Baraga, Michigan (43898)
20 -- Jan231 Monahan736 (7 y/o M) Baraga, Michigan (9871)
Records: total=24, alive=20, dead=4
RNG=20
Clinician RNG=11272

You've just generated 24 patients! We're happy to know Synthea is helping you with your project.
Like what we're doing? Synthea is an open-source project, and your support helps us keep building.
Please consider making a donation (https://github.com/sponsors/synthetichealth).
```


4. Marquette generation

```
Activities Terminal Mar 12 01:24 sat3210@uphile: ~/synthea
Consider enabling configuration cache to speed up this build: https://docs.gradle.org/5.6.1/userguide/configuration_cache.html
(base) sat3210@uphile:~/synthea$ ./run_synthea -p 20 Michigan "Marquette" --config covid19

> Task :run
SLF4J: No SLF4J providers were found.
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#noProviders for further details.
Scanned 90 modules and 157 submodules.
Loading submodule modules/allergies/allergy_panel.json
Loading submodule modules/allergies/drug_allergy_incidence.json
Loading submodule modules/allergies/environmental_allergy_incidence.json
Loading submodule modules/allergies/food_allergy_incidence.json
Loading submodule modules/allergies/immunotherapy.json
Loading submodule modules/allergies/outgrow_env_allergies.json
Loading submodule modules/allergies/outgrow_food_allergies.json
Loading submodule modules/allergies/severe_allergic_reaction.json
Loading submodule modules/anemia/anemia_sub.json
Loading submodule modules/breast_cancer/chemotherapy_breast.json
Loading submodule modules/breast_cancer/hormone_diagnosis.json
Loading submodule modules/breast_cancer/hormonotherapy_breast.json
Loading submodule modules/breast_cancer/surgery_therapy_breast.json
Loading submodule modules/breast_cancer/tnm_diagnosis.json
Loading submodule modules/contraceptives/clear_contraceptive.json
Loading submodule modules/contraceptives/female_sterilization.json
Loading submodule modules/contraceptives/implant_contraceptive.json
Loading submodule modules/contraceptives/injectable_contraceptive.json
Loading submodule modules/contraceptives/intrauterine_device.json
Loading submodule modules/contraceptives/male_sterilization.json
Loading submodule modules/contraceptives/oral_contraceptive.json
Loading submodule modules/contraceptives/patch_contraceptive.json
Loading submodule modules/contraceptives/ring_contraceptive.json
Loading submodule modules/covid19/admission.json
Loading submodule modules/covid19/determine_risk.json
Loading Lookup Table: covid-19-severity-outcomes.csv
Loading Lookup Table: covid-19-survival-outcomes.csv
Loading submodule modules/covid19/diagnose_bacterial_infection.json
Loading submodule modules/covid19/diagnose_blood_clot.json
Loading submodule modules/covid19/end_outcomes.json
Loading submodule modules/covid19/end_symptoms.json
Loading submodule modules/covid19/infection.json
Loading submodule modules/covid19/measurements_daily.json
Loading submodule modules/covid19/measurements_frequent.json
Loading submodule modules/covid19/measurements_vitals.json
Loading submodule modules/covid19/medications.json
Loading submodule modules/covid19/nonsurvivor_lab_values.json
```

```
Activities Terminal Mar 12 01:26 sat3210@uphile: ~/synthea
Seed: 1773290084319
Provider Seed: 1773290084319
Reference Time: 1773290084319
Location: Marquette, Michigan
Min Age: 0
Max Age: 140
2 -- Rick943 Baumbach677 (14 y/o M) Marquette, Michigan (19344)
1 -- Fritt267 Larkin917 (67 y/o M) Marquette, Michigan (112383)
3 -- Tashia814 Rish799 (46 y/o F) Marquette, Michigan DECEASED (62726)
3 -- Mik1234 W110dean275 Hames673 (82 y/o F) Marquette, Michigan DECEASED (114719)
4 -- Jot1660 Kris249 McKenzie376 (82 y/o F) Marquette, Michigan (118803)
4 -- Whitley172 Oberbrunner298 (12 y/o F) Marquette, Michigan (17074)
5 -- Marshall526 West559 (40 y/o M) Marquette, Michigan (55854)
6 -- Eldon28 Morissette803 (57 y/o M) Marquette, Michigan (95120)
7 -- Kassandra256 Mechelle851 Nolan344 (38 y/o F) Marquette, Michigan (52442)
8 -- Wilbert25 Rempel203 (4 y/o M) Marquette, Michigan (6836)
9 -- Jared66 Weissnat378 (39 y/o M) Marquette, Michigan (54190)
11 -- Garland107 Sauer652 (9 y/o M) Marquette, Michigan (12267)
10 -- Alishia395 Gibson10 (79 y/o F) Marquette, Michigan (110469)
12 -- Cheryl215 Jakubowski832 (54 y/o F) Marquette, Michigan (74024)
13 -- Delmy687 Page791 Price929 (83 y/o F) Marquette, Michigan (116257)
14 -- Cristobal567 Halvorson124 (66 y/o M) Marquette, Michigan DECEASED (102089)
15 -- Paz9 Nola814 Cassin499 (56 y/o F) Marquette, Michigan (79816)
16 -- Otis335 Osinski1784 (70 y/o M) Marquette, Michigan (117647)
14 -- Marvin195 Rohan584 (85 y/o M) Marquette, Michigan DECEASED (119045)
14 -- Carol737 Heaney114 (90 y/o M) Marquette, Michigan (124964)
17 -- Emmitt44 Bauch723 (38 y/o M) Marquette, Michigan (50825)
18 -- Gerard367 Klein929 (52 y/o M) Marquette, Michigan (70931)
20 -- America446 Pollich983 (15 y/o F) Marquette, Michigan (21192)
19 -- Shaquana156 Cummings51 (85 y/o F) Marquette, Michigan DECEASED (123009)
19 -- Hoa738 Kemberly945 Glover433 (59 y/o F) Marquette, Michigan DECEASED (80638)
19 -- Bobbys24 Angelica442 Friesen796 (87 y/o F) Marquette, Michigan DECEASED (142452)
19 -- Jeannette562 Cella938 Thiel172 (91 y/o F) Marquette, Michigan DECEASED (130753)
19 -- Felicidad691 Gerhold939 (79 y/o F) Marquette, Michigan DECEASED (112131)
19 -- Lorene243 Belia417 Schimmel440 (76 y/o F) Marquette, Michigan DECEASED (139600)
Waiting for threads to finish... java.util.concurrent.ThreadPoolExecutor$5@60b8aaf1[Shutting down, pool size = 1, active threads = 1, queued tasks = 0, completed tasks = 19]
19 -- Shirley182 Nicola388 Satterfield305 (102 y/o F) Marquette, Michigan (145840)
Records: total=30, alive=20, dead=10
RNG=20
Clinician RNG=11142

You've just generated 30 patients! We're happy to know Synthea is helping you with your project.
Like what we're doing? Synthea is an open-source project, and your support helps us keep building.
Please consider: https://github.com/synthea/synthea
```

B. FHIR output

```
Activities Terminal Mar 12 01:28
sat3210@uphile: ~/synthesia
(base) sat3210@uphile:~/synthesia$ ls output/fhir/
Adolph90_Brettenberg711_7b93bddd-e93e-84f4-9b39-2306eaa12bb.json
Alberto639_Bartows492_8c8bf6d0-30d0-830a-3e89-3875127bec5d.json
Alisha395_Gibson10_d0e4819c-be03-3c7b-48ee-e11f9827dcd.json
Allan198_Huel628_5c1f20b9-cd34-e5b1-e9cb-88c75717d17e.json
Alvaro283_Jones311_0cf4eca1-b0c4-73d8-67ab-794c9ab5f0e2.json
America446_Pollch983_854628bc-d089-f4ae-6f0c-fa2bb7f82d21.json
Angel97_Spencer878_689e0fc1-6578-b12d-2172-0ead98690e2b.json
Aurelio227_Mertz280_4d74daae-7c9e-e738-c2e3-59eb93ad2562.json
Babara869_Crooks415_d64cd615-8e3e-3534-5238-b1615730016d.json
Barney639_Abernathy254_78061c33-812e-e3bd-6dc2-4e5cbff483e.json
Bobby524_Angelica442_Friesen796_387f4871-4985-c4b3-be92-8d627a0e21e6.json
Brad807_Huel628_c025ae7a-a10e-1c8d-3fcd-2eeb8515bb3d.json
Carol737_Heaney114_d874c077-9c0b-f640-8f18-bd84ef0b7719.json
Chandra831_Schaden604_533d652b-5f85-bcb9-f50c-383c869e2429.json
Cherly215_Jakubowski832_899ca34a-4866-2e81-58f5-460d38dbd153.json
Christena299_Schroeder447_815fe2ab-4360-f6f1-2d69-eddbbd624150.json
Cornelius968_Williamson769_2e094457-867b-7b32-764c-18488908e38d.json
Coy949_Wintheiser220_0151ed3e-3661-9d0b-d764-f4ace4534f8a.json
Cristobal567_Halvorsen124_3930786f-0cfb-7c80-c076-cdf4586eaa20.json
Danlan46_Lubowitz58_d6dfe0de-58e9-7be4-4ff9-461ee152eed.json
Darln74_Barton704_f247670-24c2-f9e6-7b0e-6b8ff95b34bf.json
David245_Satterfield305_fb20cd04-a7fa-6b96-89ae-4726d5251f77.json
DeLyn607_Page791_Prince929_0238db01-bacf-56a3-9cae-159b5704616f.json
Derek111_Predoy1C534_5ac9a338-662a-52db-ffa3-bb8101a8b28f.json
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```

C. Completed Table

Hospital	Outbreak Level	% Used of Population	Hospital Population	COVID patients Generated
Aspirus Hospital	Low outbreak	<1%	3,000	21
Portage Hospital	Low outbreak	<1%	9,000	20
Baraga CMH	Medium outbreak	1-4%	7,000	20
Marquette	High outbreak	>5%	20,000	30

D. Reflection

The most challenging portion of this assignment was configuring the environment required to run the Synthea generator and ensuring that the correct components were submitted for the assignment. Initially, the system failed to generate patient records because the installed Java version, which was Java 11, was incompatible with the requirements used by Synthea. This required installing and configuring Java 17 before the program could run successfully. Additionally, it was initially unclear what specific outputs were needed to be included in the final submission, particularly regarding the screenshots of the generated HL7 FHIR .json files and how they corresponded to the four hospitals. After carefully reviewing the instructions and verifying the generated files in the /synthea/output/fhir directory, the process became clearer. Once the environment was properly configured and the submission requirements were understood, generating synthetic patient data for each hospital location became simpler. This assignment helped demonstrate how synthetic healthcare data can be generated to simulate disease outbreaks and how HL7 FHIR records can be used for syndromic surveillance in public health systems.